

Logic gate and logic circuit workbook

Workbook overview

This workbook isolates the Chapter 03 material that directly tests Boolean logic, logic-gate operation, truth tables, logic expressions, and logic-circuit construction. Other Hardware questions are omitted.

Year	Selected question groups
2021	6
2022	6
2023	6
2024	6
2025	6
2026	3

Recurring question patterns

- Identify or describe the operation of AND, OR, NOT, NAND, NOR and XOR gates.
- Translate a logic circuit into a Boolean / logic expression.
- Draw a logic circuit from a given expression or set of conditions.
- Complete or diagnose truth tables for expressions and circuits.
- Convert real-world binary conditions into logic expressions.

Practice 1 9618/11 M/J/2021 • Question 8

8 Tick (✓) **one** box in each row to identify the logic gate that each statement describes.

Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1					
The output is 1 only when both inputs are different					
The output is 1 only when both inputs are 0					

[3]

Practice 2 9618/12 M/J/2021 • Question 3

3 A logic expression is given:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

(a) Draw the logic circuit for the given expression.



[4]

(b) Complete the truth table for the logic expression:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

A	B	C	Working space	S
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 3 9618/13 M/J/2021 • Question 8

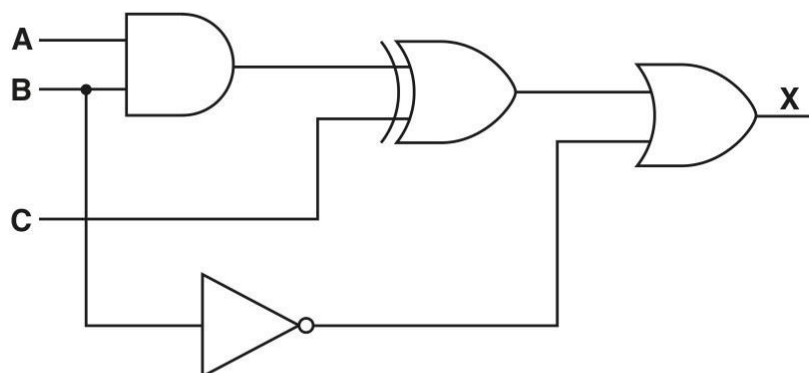
8 Tick (✓) **one** box in each row to identify the logic gate that each statement describes.

Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1					
The output is 1 only when both inputs are different					
The output is 1 only when both inputs are 0					

[3]

Practice 4 9618/11 O/N/2021 • Question 3

3 A logic circuit is shown:



(a) Write the logic expression for the logic circuit.

.....
 [3]

(b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 4 (continued)

- (c) Identify **one** logic gate **not** used in the given logic circuit. Draw the symbol for the logic gate **and** complete its truth table.

Logic gate:

Symbol:

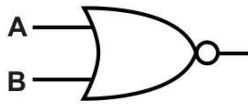
Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

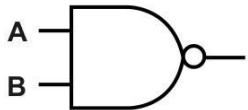
[3]

Practice 5 9618/12 O/N/2021 • Question 2

- 2 (a) Complete the truth table for each of the following two logic gates.



A	B	Output
0	0	
0	1	
1	0	
1	1	



A	B	Output
0	0	
0	1	
1	0	
1	1	

[2]

- (b) Draw a logic circuit for the following logic expression.

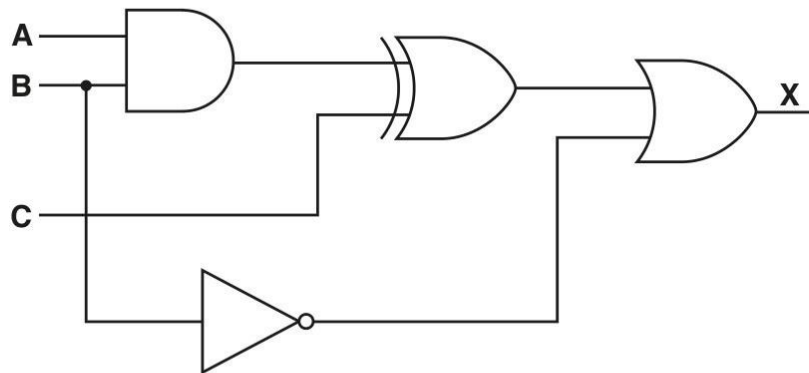
$$X = \text{NOT}(\text{NOT}(A \text{ AND } B) \text{ AND } C)$$



[2]

Practice 6 9618/13 O/N/2021 • Question 3

3 A logic circuit is shown:



(a) Write the logic expression for the logic circuit.

.....
 [3]

(b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 6 (continued)

- (c) Identify **one** logic gate **not** used in the given logic circuit. Draw the symbol for the logic gate **and** complete its truth table.

Logic gate:

Symbol:

Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

[3]

Practice 7 9618/11 M/J/2022 • Question 2(c)

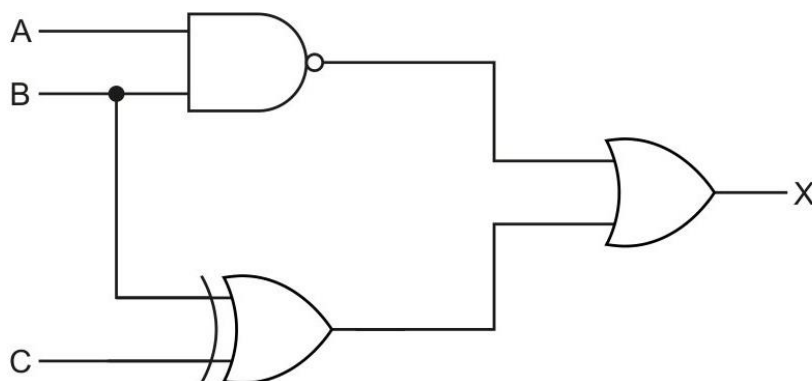
(c) Computers consist of logic gates.

- (i) Complete the table by writing **one** set of values (input 1 and input 2) for each gate that will give the output 1.

Gate	Input 1	Input 2	Output
AND			1
NAND			1
XOR			1
NOR			1

[4]

- (ii) Write the logic expression for the given logic circuit.



.....

.....

..... [3]

Practice 8 9618/12 M/J/2022 • Question 7

7 Complete the truth table for the following logic expression:

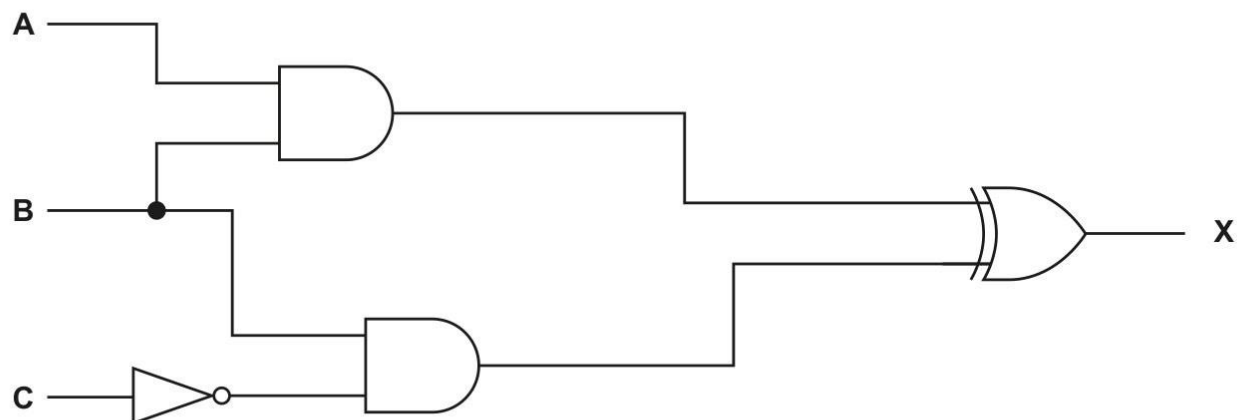
$$X = (A \text{ XOR } B) \text{ AND NOT } C$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 9 9618/13 M/J/2022 • Question 7

- 7 (a) Write the logic expression for the following logic circuit.



.....
 [3]

- (b) Complete the truth table for the following logic expression:

$$X = (A \text{ NAND } B) \text{ OR } (A \text{ AND NOT } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 10 9618/11 O/N/2022 • Question 3

- 3 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT } ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } B \text{ AND NOT } C))$$



[2]

- (b) Complete the truth table for the logic expression:

$$X = \text{NOT } ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } B \text{ AND NOT } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 11 9618/12 O/N/2022 • Question 3(a)-(b)

3 (a) A greenhouse has an automatic window.

The window (**X**) operates according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
T	Temperature	1	Too high
		0	Acceptable
W	Wind speed	1	Too high
		0	Acceptable
R	Rain	1	Detected
		0	Not detected
M	Manual override	1	On
		0	Off

The window opens (**X** = 1) if:

- the temperature is too high **and** the wind speed is acceptable
- **and**
- rain is not detected, **or** the manual override is off.

Draw a logic circuit to represent the operation of the window.



[3]

Practice 11 (continued)

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ OR } B \text{ OR } C) \text{ AND } (B \text{ NOR } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 12 9618/13 O/N/2022 • Question 5

- 5 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT } ((\text{NOT } (A \text{ AND } B)) \text{ OR } (\text{NOT } (B \text{ AND } C)))$$



[3]

- (b) Complete the truth table for the logic expression:

$$Y = (\text{NOT } P \text{ AND } Q) \text{ OR } (Q \text{ AND NOT } R)$$

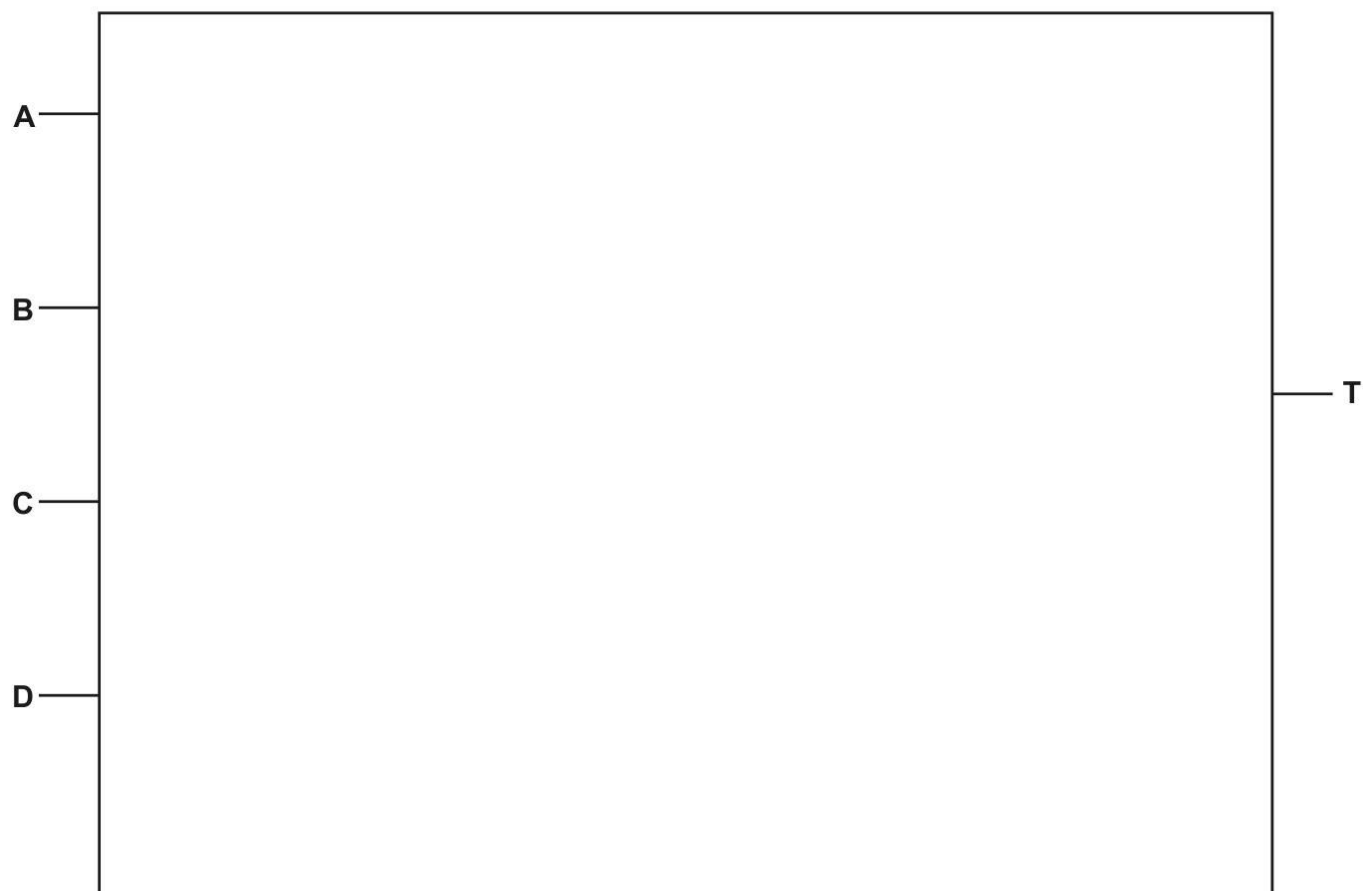
P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 13 9618/11 M/J/2023 • Question 5

- 5 (a) Draw the logic circuit for this logic expression:

$$T = (\text{NOT } A \text{ OR } B) \text{ XOR } (C \text{ NAND } D)$$



[2]

- (b) Describe the function of the NAND and NOR logic gates.

NAND

.....

NOR

.....

[2]

Practice 14 9618/12 M/J/2023 • Question 6

- 6 (a) Draw the logic circuit for this logic expression:

$$Z = (R \text{ XOR } S) \text{ AND } (\text{NOT } T \text{ NOR } P)$$



[2]

- (b) Complete the truth table for this logic expression:

$$Z = (\text{NOT } P \text{ OR } Q) \text{ XOR } (R \text{ NOR } Q)$$

P	Q	R	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

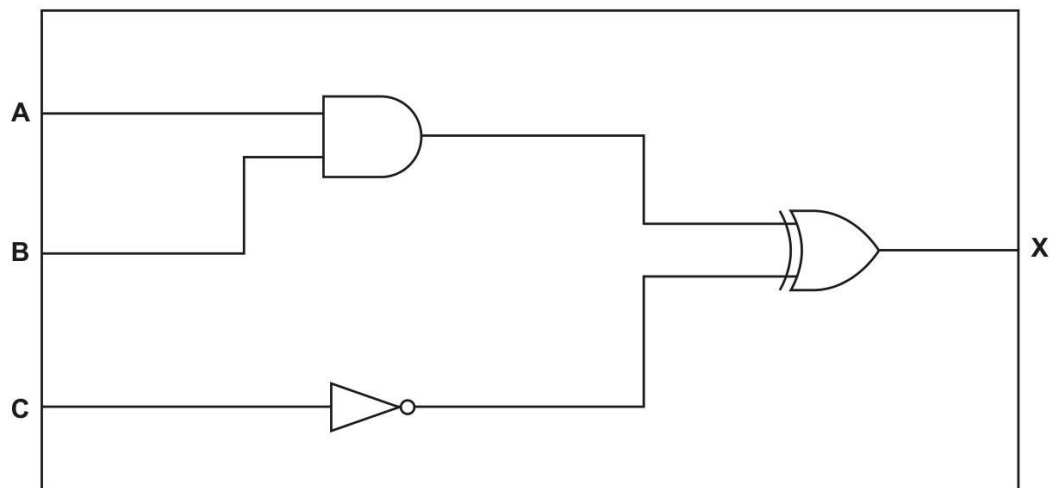
Practice 15 9618/13 M/J/2023 • Question 1

- 1 (a) Write the logic expression for this truth table:

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

..... [1]

- (b) Complete the truth table for this logic circuit:



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 16 9618/11 O/N/2023 • Question 4

- 4 (a) Complete the truth table for the logic expression:

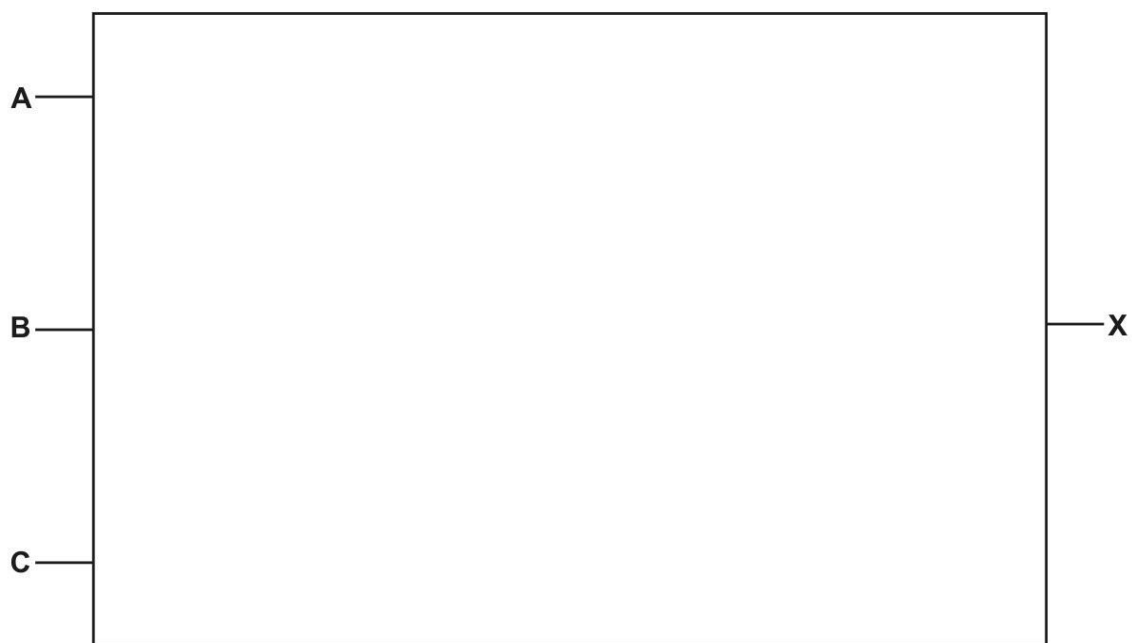
$$X = \text{NOT } (A \text{ NAND } B) \text{ XOR } (\text{NOT } B \text{ AND } (B \text{ NOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

- (b) Draw a logic circuit for the logic expression:

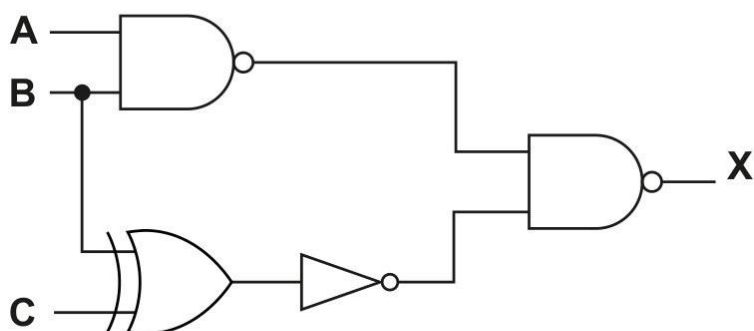
$$X = (A \text{ AND NOT } (B \text{ OR } C)) \text{ AND } (B \text{ NOR } C)$$



[2]

Practice 17 9618/12 O/N/2023 • Question 4

- 4 (a) Write the Boolean expression that corresponds to the following logic circuit.



.....
 [3]

- (b) Complete the truth table for the logic expression:

$$X = A \text{ XOR } (B \text{ AND } (A \text{ NAND } B) \text{ AND NOT } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 18 9618/13 O/N/2023 • Question 4

- 4 (a) Complete the truth table for the logic expression:

$$Y = ((P \text{ AND } Q) \text{ XOR } ((\text{NOT } Q) \text{ OR } R)) \text{ AND NOT } P$$

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

- (b) Draw a logic circuit for the logic expression:

$$Y = ((P \text{ AND } Q) \text{ XOR } ((\text{NOT } Q) \text{ OR } R)) \text{ AND NOT } P$$



[2]

Practice 19 9618/11 M/J/2024 • Question 1

- 1 (a) Tick (✓) **one** box to identify the correct logic statement for this truth table.

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

- ☐ NOT (A AND B AND C)
- ☐ (A XOR B) NOR C
- ☐ (A OR B OR C) NOR C
- ☐ NOT A AND NOT B AND NOT C

[1]

- (b) Draw a logic circuit for the logic expression:

$$X = \text{NOT} (\text{NOT } A \text{ AND } (\text{NOT } B \text{ XOR } C))$$



[2]

1 (a) Describe the operation of each of the following logic gates:

NAND

.....

NOR

.....

XOR

.....

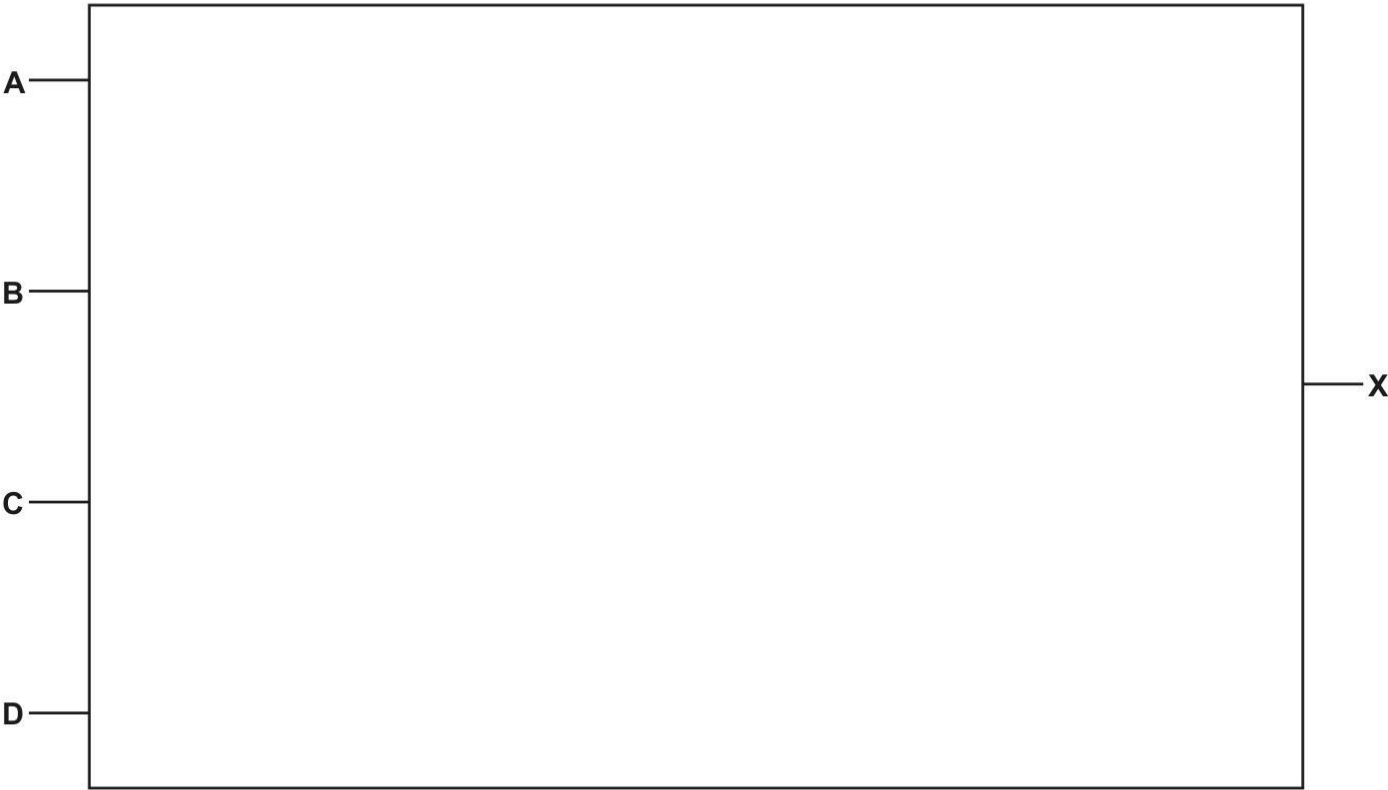
OR

.....

[4]

(b) Draw a logic circuit for this logic expression:

$$X = \text{NOT } ((A \text{ AND } B) \text{ OR } (C \text{ AND } D))$$



[2]

Practice 21 9618/13 M/J/2024 • Question 6

6 Each of the following truth tables has three inputs (**A**, **B** and **C**) and one output (**X**).

Draw **one** line to match each truth table with its logic expression.

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

Logic expression

NOT (**A** XOR **B**) AND **C**

(**A** OR **C**) AND NOT **B**

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A NAND **B** NAND **C**

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

(**A** NAND **B**) OR **C**

NOT (**A** AND **B** AND **C**)

[3]



(b) The floodlight (X) and audio alarm (Y) operate according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
A	external daylight level	1	Low
		0	High
B	front door	1	Open
		0	Closed
C	person is within 2 m	1	Detected
		0	Not detected
D	back door	1	Open
		0	Closed
E	security system	1	Switched on
		0	Switched off

The floodlight turns on (**X = 1**) if:

- the security system is switched on
and
- the external daylight level is low
and
- a person is detected within 2 m.

The audio alarm turns on (**Y = 1**) if:

- the security system is switched on
and
- one **or** more doors are open, **or** a person is detected within 2 m.

Write logic expressions for the security system.

X =

.....

Y =

.....

[2]

.....

1 (a) Complete the truth table for the logic expression:

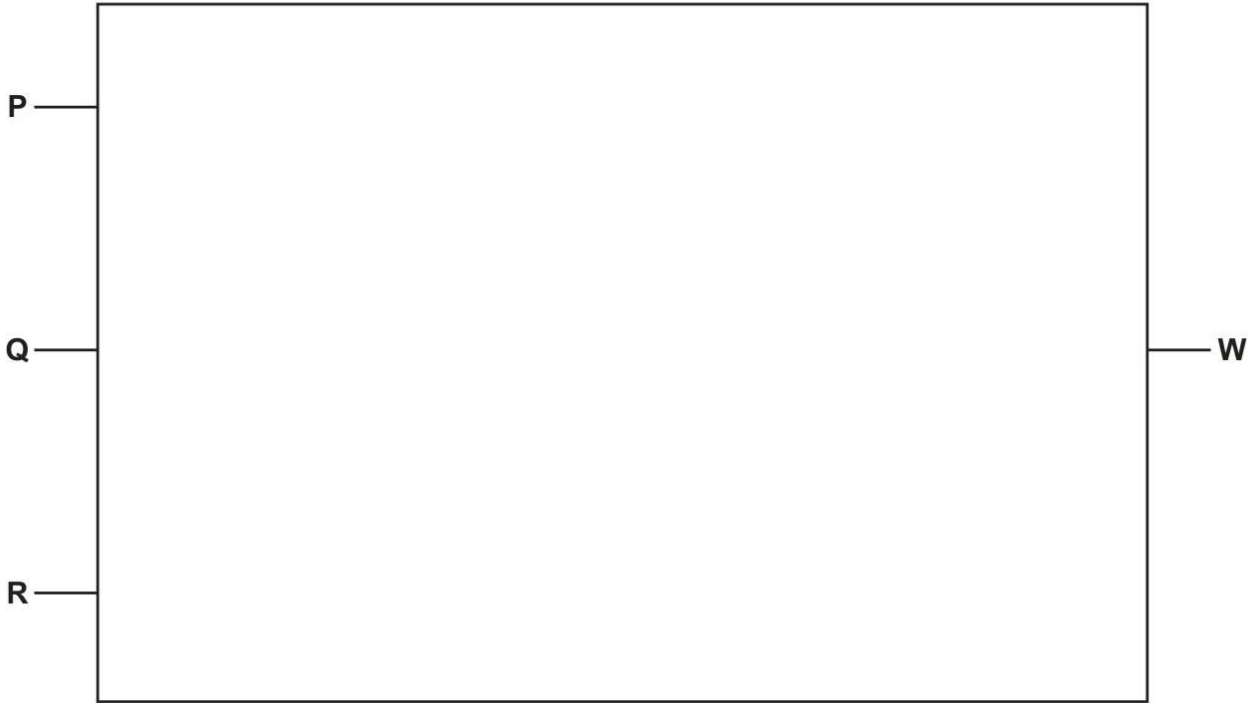
$$X = (A \text{ XOR } B) \text{ NAND } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw the logic circuit for the logic expression:

$$W = P \text{ NAND } ((Q \text{ OR NOT } R) \text{ XOR } (P \text{ XOR } Q))$$



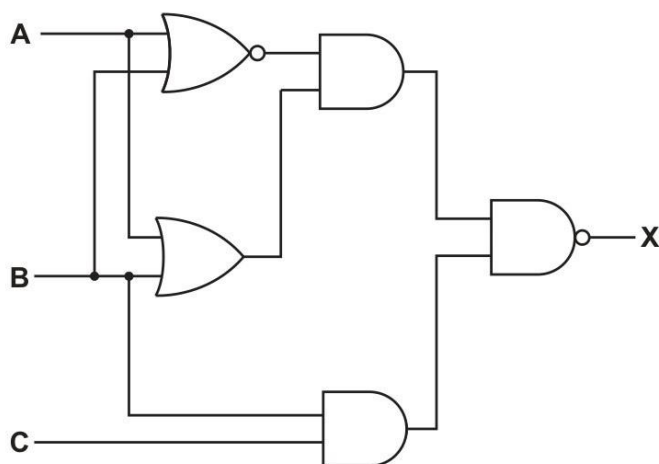
[2]

Practice 24 9618/13 O/N/2024 • Question 1

- 1 (a) Describe the operation of a 2-input XOR gate.

.....
 [1]

- (b) Write the logic expression for the following logic circuit.



$X =$
 [2]

- (c) Draw the logic circuit for the logic expression:

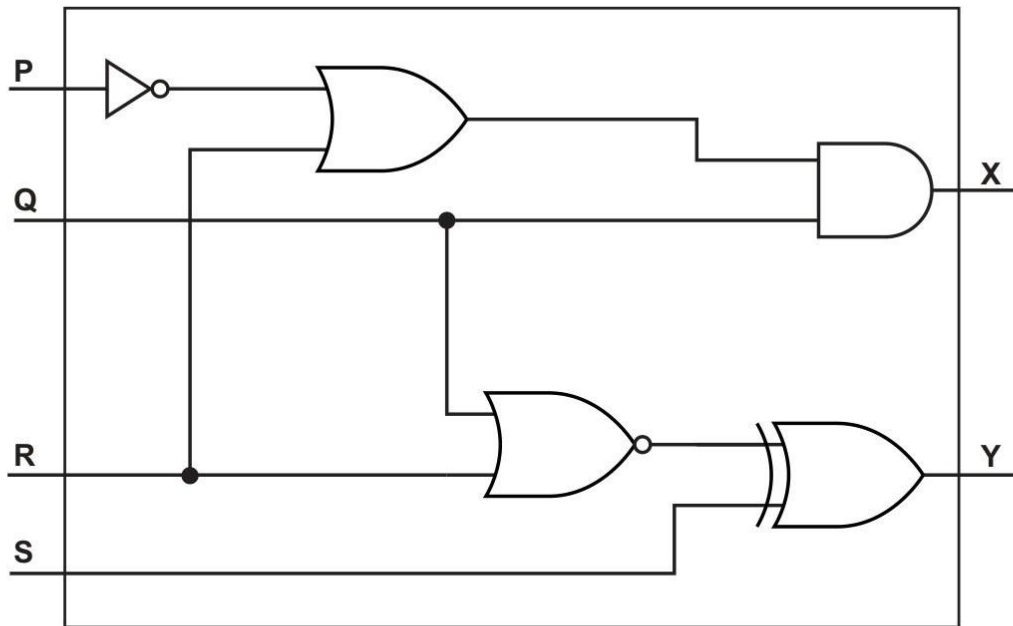
$$X = ((\text{NOT } A \text{ AND } (B \text{ AND } C)) \text{ OR } (B \text{ NAND } C)) \text{ AND NOT } A$$



[2]

Practice 25 9618/11 M/J/2025 • Question 1

- 1 (a) Write the logic expressions for the following logic circuit.



X =

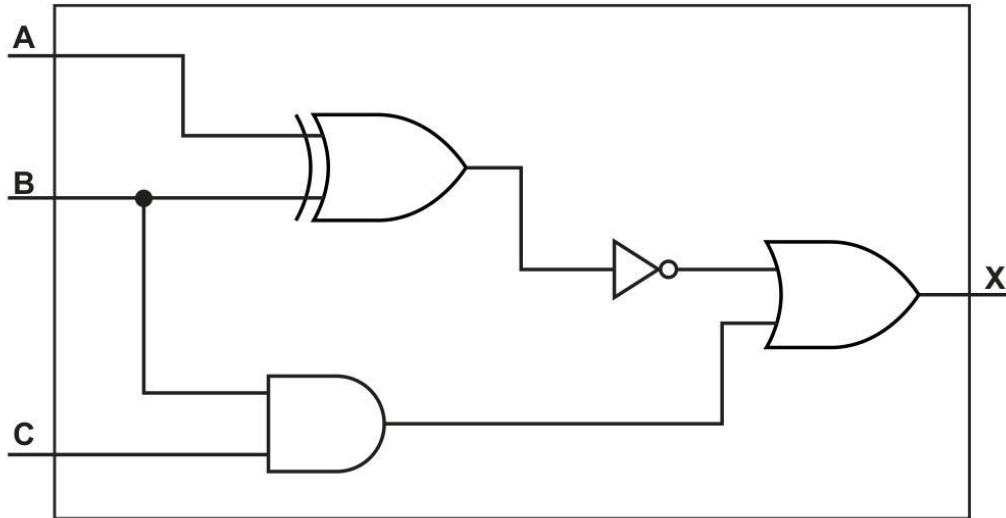
Y =

[2]

Practice 25 (continued)



(b) Complete the truth table for the following logic circuit.

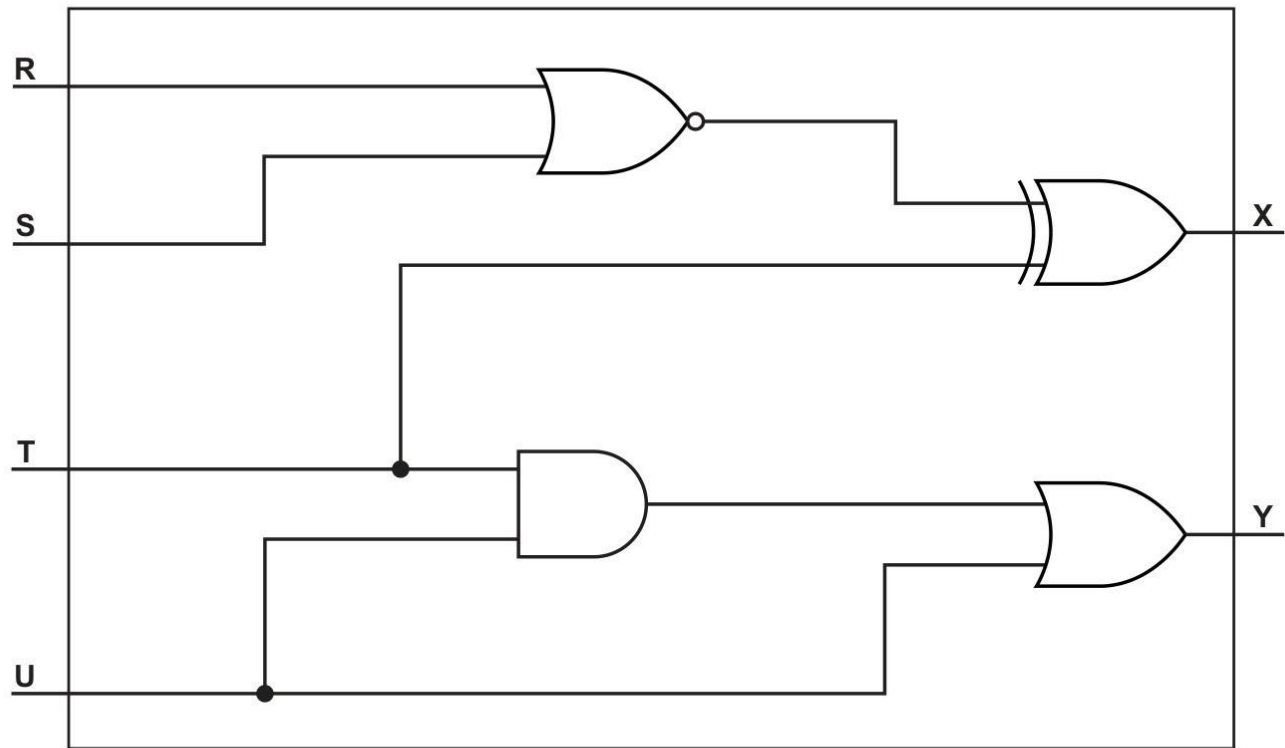


A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 26 9618/12 M/J/2025 • Question 4

- 4 (a) Write the logic expressions for the following logic circuit.



X =

Y =

[2]

Practice 26 (continued)



(b) Complete the truth table for the logic expression:

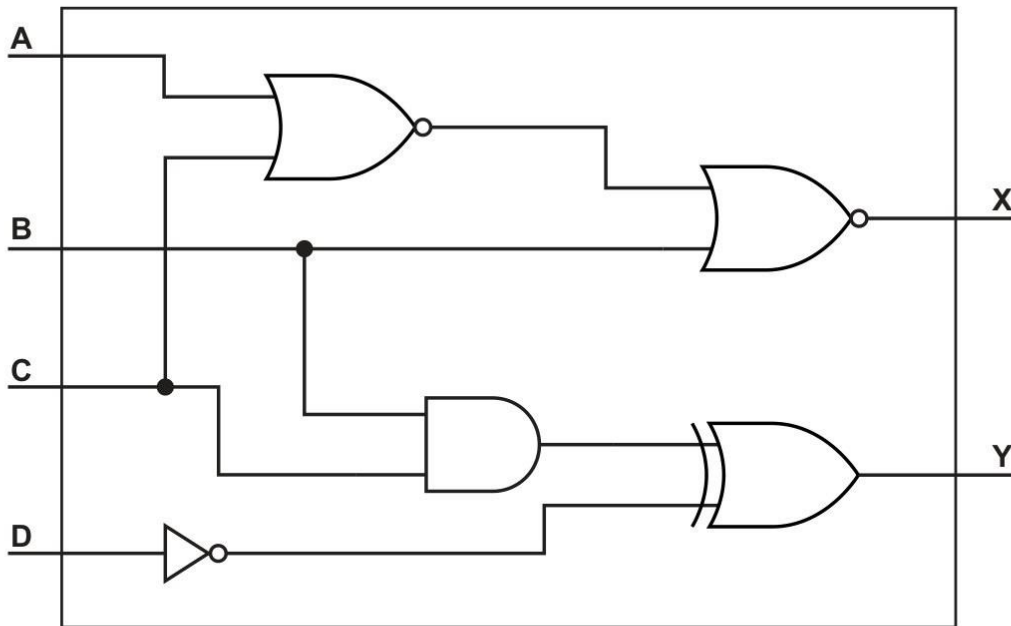
$$X = \text{NOT } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 27 9618/13 M/J/2025 • Question 4

- 4 (a) Write the logic expressions for the following logic circuit.



X =

Y =

[2]

Practice 27 (continued)



(b) Complete the truth table for the logic expression:

$$X = (A \text{ OR } B) \text{ XOR } (B \text{ OR } C) \text{ XOR } (\text{NOT } A \text{ NAND } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Practice 28 9618/11 O/N/2025 • Question 3

- 3 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT} ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$



[2]

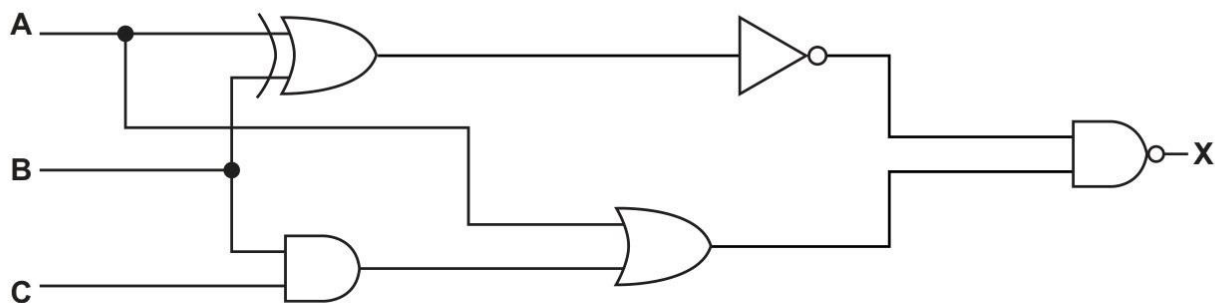
- (b) Write the logic expression for the following truth table.

R	S	T	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Q = [2]

Practice 29 9618/12 O/N/2025 • Question 9

9 (a) Consider the following logic circuit:



Write the logic expression for the logic circuit. Do **not** simplify the expression.

$X =$ [2]

(b) Consider the following logic expression:

$$X = \text{NOT } ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$

A truth table for the logic expression is given:

Row number	A	B	C	X
1	0	0	0	1
2	0	0	1	0
3	0	1	0	1
4	0	1	1	0
5	1	0	0	1
6	1	0	1	0
7	1	1	0	1
8	1	1	1	1

There are **three** errors in the truth table.

Identify the **three** errors in the truth table by writing the row numbers with an incorrect output.

Error 1 Row number

Error 2 Row number

Error 3 Row number

[2]

Practice 30 9618/13 O/N/2025 • Question 3

- 3 (a) Complete the truth table for the following logic expression:

$$X = (A \text{ NOR } B) \text{ NAND } (C \text{ XOR } B)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

- (b) Draw a logic circuit for the logic expression:

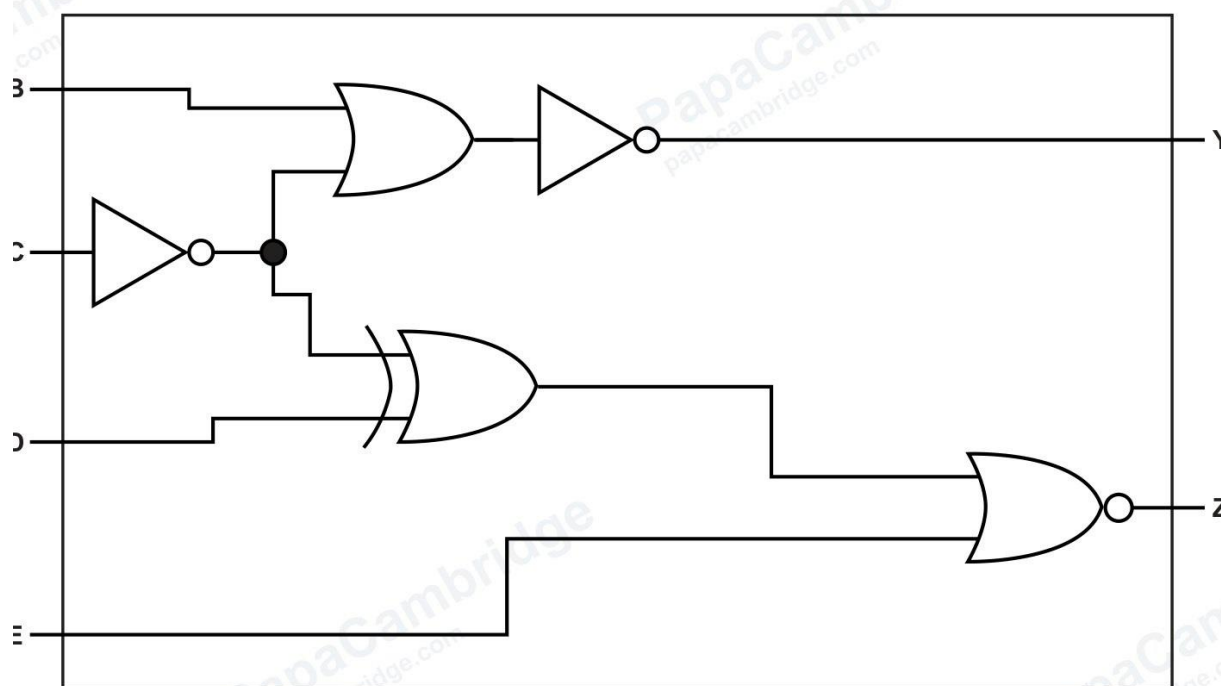
$$X = \text{NOT } (((A \text{ AND } B) \text{ OR } C) \text{ AND } (C \text{ NOR } D))$$



[2]

Practice 31 9618/11 M/J/2026 • Question 6

6 (a) Write the logic expressions for the following logic circuit.



Y =

Z =

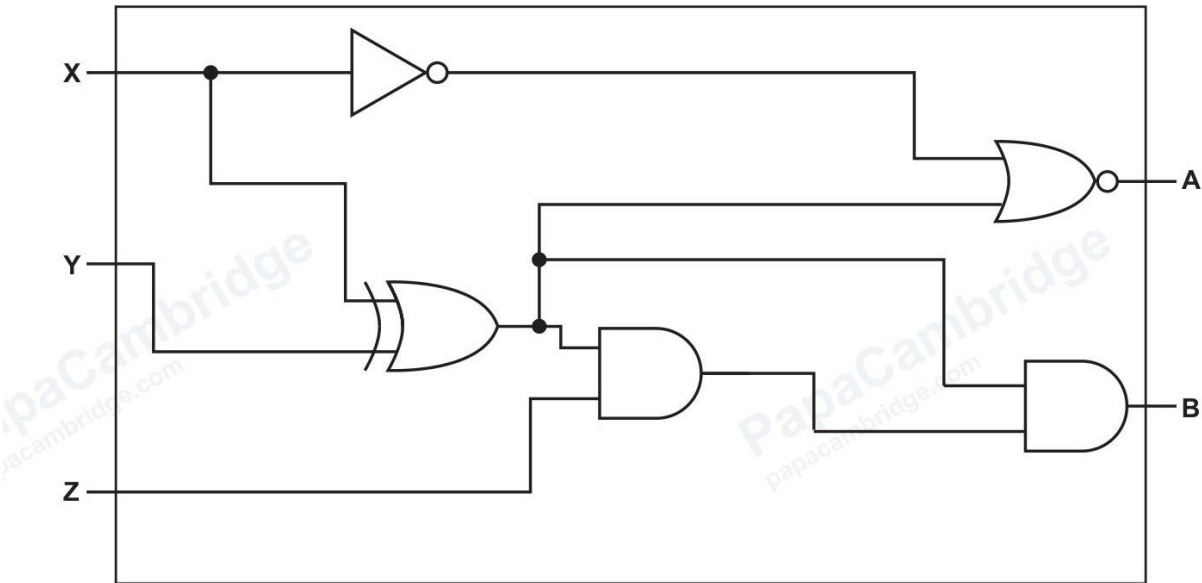
[2]

(b) Create and complete the truth table for a two-input NAND gate.

[2]

Practice 32 9618/12 M/J/2026 • Question 8

8 (a) Complete the truth table for the given logic circuit.



X	Y	Z	working space	A	B
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

[2]

Practice 32 (continued)

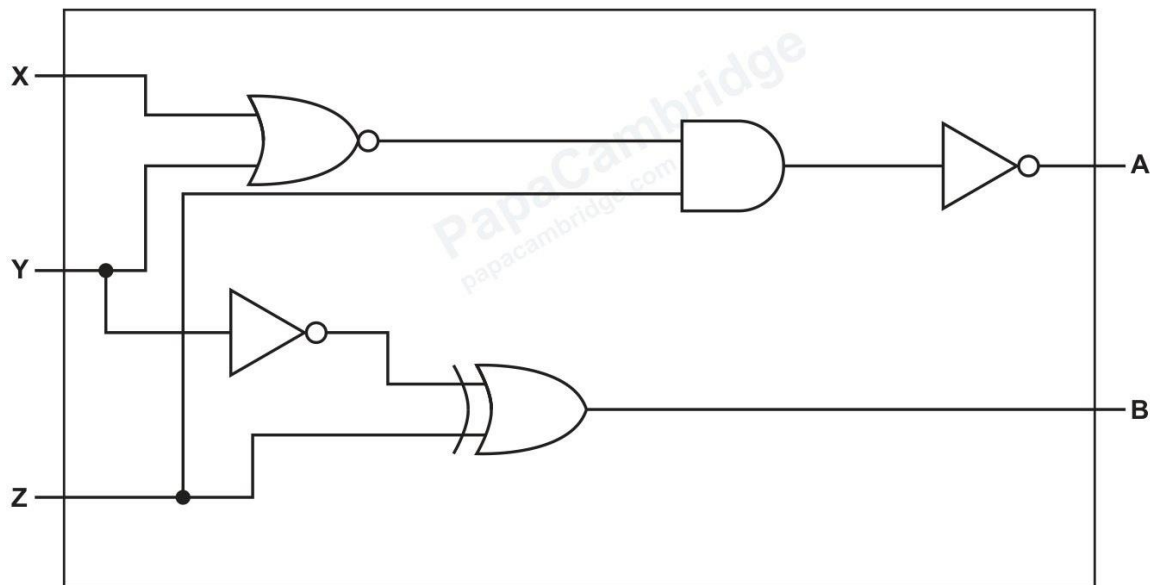
(b) Write the logic statement for the given truth table.

input			output
A	B	C	P
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

P = [1]

Practice 33 9618/13 M/J/2026 • Question 2

2 (a) Complete the truth table for the given logic circuit.



X	Y	Z	working space	A	B
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

[2]

(b) State **two** logic gates that are not in the circuit in 2(a).

..... [1]